

Activity 3: Exploring Tokens, Context Windows, and Temperature

Purpose

Large Language Models (LLMs) generate responses by processing text as tokens and maintaining conversation context within a limited context window. The quality and style of responses are also influenced by parameters such as temperature. In this activity, you will explore these concepts through practical experiments and observe how they affect AI-generated outputs.

Pre-requisites

- Completion of Activities 1 and 2
 - ChatGPT or Oracle Chat AI
 - Internet connection
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Estimated Duration

35 Minutes

Scenario

Your development team has started using AI extensively for requirement analysis and coding assistance. During a discussion, your Team Lead asks:

- Why does AI sometimes forget earlier information?
- Why do repeated prompts sometimes produce different answers?
- What are tokens?
- Why do AI responses vary?

To answer these questions, you will perform a series of experiments using Generative AI.

Tasks

Complete the following tasks in sequence.

Task 1: Observe How AI Uses Context

Start a new conversation.

Prompt

Remember the following project information.

Project Name: SmartBank

Modules:

- Customer Login
- Dashboard
- Fund Transfer
- Beneficiary Management
- Bill Payments
- Transaction History

Reply only with:

"I have remembered the project information."

After AI responds, continue with:

List all modules of the SmartBank application that I shared earlier.

Observe whether AI correctly remembers the information.

Task 2: Test Context Retention

Continue the same conversation.

Prompt

Now add the following modules to the project.

- Credit Card Services
- Loan Management
- Investment Portfolio
- Customer Support

Next, ask:

List every module currently included in the SmartBank application.

Observe whether AI combines both sets of modules correctly.

Task 3: Start a New Conversation

Open a **new chat window**.

Without providing any previous context, ask:

List all SmartBank modules we discussed earlier.

Compare the response with the previous conversation.

Task 4: Explore Temperature

Run the following prompt multiple times.

Suggest five modern names for a Net Banking Application.

Observe:

- Are all responses identical?
- Which names are more creative?
- Which responses are more consistent?

Note: If your AI platform allows temperature adjustment, repeat this activity using Low, Medium, and High temperature settings and compare the results.

Task 5: Generate Creative vs Deterministic Outputs

Prompt A

Generate a professional welcome message for users logging into the SmartBank application.

Prompt B

Generate a creative and engaging welcome message for users logging into the SmartBank application.

Compare both responses.

Task 6: Explore Token Usage

Use the following prompt.

Summarize the SmartBank project in one sentence.

Now continue with:

Summarize the SmartBank project in one paragraph.

Finally, ask:

Provide a detailed explanation of the SmartBank project in approximately 500 words.

Compare all three outputs.

Discuss:

- Which response is shortest?
- Which response contains the greatest level of detail?
- How does increasing the response length affect the number of tokens used?

Task 7: Experiment with Response Constraints

Prompt

Explain the Fund Transfer module.

Provide your answer in:

- Exactly 30 words

Then,

- Exactly 100 words

Finally,

- A detailed explanation using bullet points.

Observe how the level of detail changes with the requested output length.

Expected Deliverables

Upon completion of this activity, you should have:

- Understanding of conversation context
- Understanding of context windows
- Examples of AI remembering previous conversations
- Examples of AI losing context in a new conversation

- Comparison of creative vs deterministic responses
 - Comparison of short and long responses
 - Understanding of how response length influences token usage
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Challenge Task

Imagine you are preparing AI guidelines for new software engineers joining your organization.

Use AI to explain the following concepts in simple language suitable for beginners.

Prompt

Create a beginner-friendly guide explaining the following AI concepts.

- Tokens
- Context Window
- Temperature

For each concept include:

- Simple Definition
- Real-world Example
- Why Software Engineers Should Understand It

Present the information using headings and bullet points.